

Collaboration Sociometry — Scoring Card

A peer-network instrument. Every member of an intact group rates every other member on twelve statements. The unit of analysis is the cohort, not the person.

THE QUESTION THIS INSTRUMENT ANSWERS FIRST

Most trusted

Trust block – items 8, 9, 10

The share of a person’s raters who find them reliable, safe to be wrong in front of, and straight with difficult news.

Most power — over

Power – over block – items 5, 6, 7

The share of raters who adjust to this person’s position, treat their approval as consequential, or see them shape the agenda informally.

Most power — to & with

Power – to & with block – items 1, 2, 3, 4

The share of raters who seek this person’s judgement, depend on them to unlock resources, or route work through them.

All three are the same computation run on a different block of statements. Nothing in this instrument produces an overall "top person": a name leads the Trust ranking or the Power ranking, never the cohort as a whole.

HOW A RANKING IS PRODUCED

01	A tie, not a score	Ranking is not done on the average. For each colleague who rated this person, their row mean for the block is taken; if it is at or above the tie threshold (4) that colleague has drawn a tie to them.
02	Tie rate, not tie count	The person is then ranked on the share of their raters who drew a tie — ties ÷ raters — rather than the raw count. Without this, whoever is known by the most people wins by default.
03	Ordering	Highest tie rate first. Where two people tie exactly, the higher block mean comes first; where those are equal too, roster order decides, which is arbitrary and should not be read as a ranking.
04	Read per block	The same procedure on the Trust block answers "most trusted"; on Power — over it answers "who is most deferred to"; on Power — to & with, "who enables and connects most".

Coverage is not part of the ranking

A tie rate says nothing about how many people it rests on. In the worked example below, Eve was rated by a single colleague who rated her highly, giving her a tie rate of 1.00 — joint top on all four networks, and first outright on Power — to & with, above three people whose whole working group rated them. Her coverage of 1 is below the rater floor of 3, so the platform withholds her individual profile, but the ranked lists are not filtered on the same rule. Read every ranking beside the rater count, and treat a thinly-rated name near the top as an artefact of coverage rather than a finding about the group.

HOW A RATING BECOMES A SCORE

01	Rate	Each respondent rates each assigned colleague on 12 statements, 1–5. A colleague may be left entirely blank — that is a valid answer, meaning "no basis to judge".
02	Row mean	For one rater and one target, average that rater's answers within a block.
03	Block mean	Average those row means across every rater who rated the target. Two steps, not one: a rater who answered a block counts once, whatever the block contains.
04	Band and tie	Band the block mean. Separately, count each row mean at or above the tie threshold (4) — that count is the person's in-degree in that block's network.
05	Aggregate	Roll the per-member figures into group density, reciprocity, concentration and the group signals.

Averaging every raw answer in one pass would weight a rater by how many statements they happened to answer. Averaging each rater's row first, then averaging the rows, gives every rater one vote per block. Partial rows cannot arise in practice — the platform refuses a submission where a colleague is part-rated, at the browser and again at the API — but the two-step mean is what makes the figure well-defined regardless.

CONFIGURATION

Rating scale 1–5, fixed. Anchors: Not at all true · Slightly true · Moderately true · Mostly true · Completely true.

Tie threshold	Default 4 ("Mostly true") — the first point where a respondent asserts the statement rather than conceding part of it. Configurable per cohort, 1–5.
Rater floor (minRaters)	Default 3. Below it a member gets no individual profile. Configurable per cohort, 1–50.
Minimum rated targets	Default 1. How many colleagues a respondent must rate before submitting. Configurable per cohort, 0–200.
Reports to participants	Off by default. Turned on per cohort.
Email OTP	Off by default. When on, a one-time code must verify the roster email before the exercise opens.

THE TWELVE STATEMENTS

Verbatim from the client workbook's Instructions sheet, in the workbook's own column order.

NO	SHORT LABEL	STATEMENT	BLOCK
1	Trusted judgment	When I face a difficult problem, I seek out this person's judgment, and I find myself adopting their view out of the respect I have for them.	Power — to & with
2	Unlocks resources	This person can unlock resources, budget, or priority that I depend on.	Power — to & with
3	Rallies others	This person brings people together and builds shared commitment across teams.	Power — to & with
4	Route through	To get things moving across departments, this is the person I route them through — and the one I rely on to know what's really happening across the organisation.	Power — to & with
5	Others fall in line	When this person takes a firm position, others here tend to adjust and fall in line with it.	Power — over
6	Approves or blocks	This person can approve or hold up an initiative largely on their own, and their backing or disapproval has real consequences for how things go for people.	Power — over
7	Shapes it informally	This person shapes which issues get attention, often informally and before they reach the room.	Power — over
8	Reliable delivery	I get what I need from this person on time, as promised, and I trust the quality of their work enough to build on it without re-checking.	Trust
9	Looks out for me	This person looks out for my interests and the shared goal, and I could admit a mistake or ask them for help without fear it would be used against me.	Trust
10	Keeps their word	This person does what they say they will, even when it is inconvenient, and shares information with me openly, including difficult news, early.	Trust
11	Easy to work with	Working with this person is straightforward and productive.	Overall ease
12	Wish for more support	I would like more support or cooperation from this person in my work than I currently get.	Support gap — scored alone

TWO ITEMS HANDLED UNUSUALLY

Item 12 — deficit polarity	The only statement where a high score is a complaint rather than a strength. It is kept out of all four blocks and out of the cohort mean, and banded on its own scale. It is not reverse-scored: no answer anywhere in this instrument is recoded.
Item 10 — double-barrelled	It asserts two things at once — keeping one's word, and sharing difficult news early. It counts in full toward the Trust block, but is excluded from the reliability/openness facet split (items 8 and 9), because assigning it to either facet would invent a reading the respondent never gave. Pending a client decision on splitting the item.

THE FOUR BLOCKS

BLOCK	ITEMS	READS
● Power — to & with	1, 2, 3, 4	who enables and connects
● Power — over	5, 6, 7	who controls and decides
● Trust	8, 9, 10	how you experience working with them
● Overall ease	11	straightforward and productive to work with
● Support gap	12	where the group asks for more than it gets — never averaged into a block

BANDS — ASSET BLOCKS

Low	$\text{mean} < 2.50$	The group does not, on the whole, see this in the person.
Mixed	$2.50 \leq \text{mean} < 3.50$	Divided, or moderate across the board.
Strong	$3.50 \leq \text{mean} < 4.25$	Consistently seen.
Very strong	$\text{mean} \geq 4.25$	Close to unanimous, near the top of the scale.

BANDS — SUPPORT GAP (ITEM 12)

Little	$\text{mean} < 2.50$	The group is broadly getting what it needs from this person.
Some	$2.50 \leq \text{mean} < 3.50$	A real, moderate ask for more.
Marked	$\text{mean} \geq 3.50$	A clear and widely held request for more support.

Bands are read on the block mean only. A high band on a thinly rated person is still thinly rated — every figure the platform reports carries its rater count (n) beside it, and below the rater floor no profile is produced at all.

EVERY FIGURE THE ENGINE PRODUCES

What a member receives

coverage	How many colleagues rated this person.	The denominator behind every figure in their profile.
possibleRaters	Respondents, excluding the person themselves.	What coverage could have been.
suppressed	$\text{coverage} < \text{minRaters}$	True means no individual profile is produced at all.
items[].mean	Mean of one statement across the raters who answered it.	Carries its own n.
blocks[].mean	Mean of per-rater row means, across raters.	The two-step mean. Rounded to 2dp.
blocks[].band	The band that mean falls in.	Low / Mixed / Strong / Very strong.
blocks[].ties	Raters whose row mean was at or above the threshold.	The person's in-degree in that block.
blocks[].tieRate	$\text{ties} \div n$	Coverage-normalised, so a person rated by 4 compares with one rated by 12.
supportGap	Item 12 alone, with its own band.	Never averaged into a block.
authorityTrustGap	$\text{power_over mean} - \text{trust mean}$	Positive: complied with more than relied on. Negative: relied on more than deferred to. Null when either side has no ratings.

What a member gives

given.outDegree	How many colleagues this person rated.	A measure of reach, not of quality.
given.mean	Mean of every asset-polarity rating they gave.	Item 12 excluded.
given.versusCohort	$\text{given.mean} - \text{cohortMean}$	Leniency, not accuracy. It says how this rater scores relative to the group, never whether they are right.
responded	Whether this member submitted a response of their own.	A member can be rated without responding, and vice versa.

The group, per block

ties	Directed ties at or above the threshold in this block.	One per rater→target row that cleared the bar.
density	$\text{ties} \div \text{rated pairs}$	How much of the working contact in the group carries this kind of tie.
reciprocity	Mutual pairs carrying both ties $\div$ mutual pairs carrying at least one.	Counted only on pairs who actually rated each other, so two people who simply do not work together never read as coldness.
concentration	Summed shortfall from the highest tie rate $\div$ the largest shortfall a group this size could produce.	0 = flat, 1 = one person holds every tie. Built on coverage-normalised tie rates. Null below two members.
ranked	Members ordered by tie rate, then mean, then roster position.	Most-connected first.

Group signals

responseRate	Respondents $\div$ roster size.	
acquaintance	Ratings given $\div$ pairs those respondents could have rated.	How much of the group knows itself well enough to judge.
cohortMean	Mean of every asset-polarity rating given by anyone to anyone.	The baseline for leniency. Item 12 excluded.
isolates	Members rated by colleagues, yet carrying zero ties in all four blocks.	A finding about connection, not a low score. A member nobody rated is not an isolate.
underCovered	Members below the rater floor.	No individual report is produced for them.
authorityWithoutTrust	The five largest positive authority-trust gaps.	Complied with, more than relied on. Suppressed members are excluded.
trustWithoutAuthority	The five largest negative gaps.	Relied on, more than deferred to.
supportGaps	The five highest item-12 means.	Where the group asks for more than it gets.
functionMatrix	Mean trust rating from each function to each function.	The seams between departments. Blank where no pair was rated.

A SIX-PERSON COHORT, WORKED END TO END

Built to exercise every mechanic above. Every figure was produced by the live scoring engine when this document was built — none of it is typed in.

The roster

MEMBER	FUNCTION	RESPONDED	RATERS	SUPPRESSED
Alice	Operations	yes	3	no
Bob	Quality	yes	3	no
Cara	R&D	yes	3	no
Dan	IT	yes	3	no
Eve	Finance	no	1	yes
Frank	Operations	no	0	yes

The ratings that were given

TARGET	POWER TO/WITH	POWER OVER	TRUST	EASE	ITEM 12	RATED BY
Alice	4	5	2	3	4	Bob, Cara, Dan
Bob	4	2	5	5	1	Alice, Cara, Dan
Cara	4	4	4	3	2	Alice, Bob, Dan
Dan	2	2	2	2	5	Alice, Bob, Cara
Eve	5	5	5	5	1	Alice only
Frank	—	—	—	—	—	nobody

Every rater gave the same figures for a given colleague, so the arithmetic stays checkable by hand. Real data never looks like this — the spread between raters is itself a finding, and the platform reports each figure with the rater count behind it.

Block means, bands and ties

MEMBER	POWER TO/WITH	POWER OVER	TRUST	EASE	SUPPORT GAP	AUTHORITY – TRUST
Alice	4.00 Strong · 3/3	5.00 Very strong · 3/3	2.00 Low · 0/3	3.00 Mixed · 0/3	4.00 Marked	+3.00
Bob	4.00 Strong · 3/3	2.00 Low · 0/3	5.00 Very strong · 3/3	5.00 Very strong · 3/3	1.00 Little	-3.00
Cara	4.00 Strong · 3/3	4.00 Strong · 3/3	4.00 Strong · 3/3	3.00 Mixed · 0/3	2.00 Little	0.00
Dan	2.00 Low · 0/3	2.00 Low · 0/3	2.00 Low · 0/3	2.00 Low · 0/3	5.00 Marked	0.00
Eve	5.00 Very strong · 1/1	5.00 Very strong · 1/1	5.00 Very strong · 1/1	5.00 Very strong · 1/1	1.00 Little	0.00
Frank	—	—	—	—	—	—

Worked by hand — Alice’s authority-trust gap

Who rated Alice Bob, Cara and Dan. Alice cannot rate herself; Eve and Frank did not respond.

Each rater’s Power-over row mean Items 5, 6 and 7 are all 5, so every row mean is 5.00.

Power-over block mean	Three row means of 5.00 average to 5.00 → band Very strong. All three clear the threshold of 4, so ties = 3 of 3.
Each rater's Trust row mean	Items 8, 9 and 10 are all 2, so every row mean is 2.00.
Trust block mean	2.00 → band Low. None of the three reaches the threshold, so ties = 0 of 3.
The gap	5.00 – 2.00 = 3.00. Positive, and the largest in the cohort, so Alice heads the "complied with, more than relied on" list.
How to read it	The group adjusts to Alice when she takes a position, but does not rely on her or feel safe with her to the same degree. That is a finding about the group's relationship to her authority — not a verdict on her competence.

THE RANKINGS THIS COHORT PRODUCED

Ordered by tie rate. The rater count is printed beside every figure on purpose — read the two together.

Power — to & with

	MEMBER	TIE RATE	TIES / RATERS	BLOCK MEAN
1	Eve below rater floor	100%	1/1	5.00
2	Alice	100%	3/3	4.00
3	Bob	100%	3/3	4.00
4	Cara	100%	3/3	4.00
5	Dan	0%	0/3	2.00

Power — over

	MEMBER	TIE RATE	TIES / RATERS	BLOCK MEAN
1	Alice	100%	3/3	5.00
2	Eve below rater floor	100%	1/1	5.00
3	Cara	100%	3/3	4.00
4	Bob	0%	0/3	2.00
5	Dan	0%	0/3	2.00

Trust

	MEMBER	TIE RATE	TIES / RATERS	BLOCK MEAN
1	Bob	100%	3/3	5.00
2	Eve below rater floor	100%	1/1	5.00
3	Cara	100%	3/3	4.00
4	Alice	0%	0/3	2.00
5	Dan	0%	0/3	2.00

Overall ease

	MEMBER	TIE RATE	TIES / RATERS	BLOCK MEAN
1	Bob	100%	3/3	5.00
2	Eve below rater floor	100%	1/1	5.00
3	Alice	0%	0/3	3.00
4	Cara	0%	0/3	3.00
5	Dan	0%	0/3	2.00

What the group signals picked up

Complied with, more than relied on	Alice (+3.00)	Power-over sits above trust.
Relied on, more than deferred to	Bob (−3.00)	Trusted, without the leverage to act.
Isolates	Dan (rated by 3)	Rated by colleagues, yet tied to by none of them in any block. A finding about connection, not a low score.
Under-covered	Eve (1), Frank (0)	Below the rater floor of 3, so no individual profile is produced.
No profile at all	Frank	Nobody rated them. Not an isolate — the engine distinguishes "unrated" from "rated but untied".
Largest support gaps	Dan (5.00), Alice (4.00), Cara (2.00), Bob (1.00)	Where the group asks for more than it currently gets.

Network figures

BLOCK	TIES	DENSITY	RECIPROCITY	CONCENTRATION
Power — to & with	10	0.77	0.50	0.25
Power — over	7	0.54	0.20	0.50
Trust	7	0.54	0.20	0.50
Overall ease	4	0.31	0.00	0.75

Cohort figures

Roster size	6	Six people on the roster.
Respondents	4	Alice, Bob, Cara and Dan submitted; Eve and Frank did not.
Response rate	0.67	Respondents over roster size.
Ratings given	13	Rater→target rows actually completed.
Possible pairs	20	What those four respondents could have rated.
Acquaintance	0.65	How much of the group knows itself.
Cohort mean	3.47	Baseline for leniency, asset items only.
Rater floor	3	The platform default. Configurable per cohort.
Tie threshold	4	The default.

RULES THE ENGINE WILL NOT BEND

Enforced in code, not by convention. Each is covered by a unit test.

Blanks are data	A blank is never imputed, never zero-filled, never quietly averaged over. "We don't really work together" is the instrument's own instruction, and it is what makes the coverage figures mean anything. A blank is stored as the absence of a row, not as a sentinel inside the 1–5 range.
A started colleague must be finished	Every statement about a person is mandatory once that person is rated at all — two people rated on different subsets of the statements are not comparable numbers, however alike a report makes them look. Enforced in the browser and again at the API, which rejects a part-rated submission by name.
Self-ratings are dropped	The instrument forbids them; a self-rating that reached the API is discarded rather than scored.

Ratings off the roster are dropped	A rating aimed at a roster position that no longer exists is discarded. The roster is the authority on who is in the cohort.
Out-of-range values are rejected	A value outside 1–5 raises an error. It is never clamped into the scale, because a clamp would turn a data fault into a plausible-looking number.
Nothing is reverse-scored	No answer is recoded anywhere in this instrument. The one deficit-polarity item is handled by keeping it out of the blocks, not by flipping it.
CONFIDENTIALITY	
No individual attribution	The group report carries no figure traceable to one respondent. Who rated whom is never shown, and no report names a rater.
The rater floor	Below 3 raters (the default; configurable per cohort) no individual profile is produced at all. In a small named group a two-rater average is a quotation with a decimal point on it.
Reports need a quorum	A cohort round produces no report until at least the rater floor has responded, so that no figure can be traced back to a single person.
Before identification	The API sends no roster names at all until a respondent has identified themselves against the roster. An unenrolled email gets a plain "contact the cohort admin".
WHAT THE ENGINE DELIBERATELY DOES NOT PRODUCE	
No composite score	There is no single number per person and no overall ranking of people. Every figure is a block mean or a network statistic, and every one carries the rater count behind it.
No inference about a rater	A rater's leniency figure says how they score relative to the group. It never says whether they are right.
No imputation of the unknown	Where a pair did not work together, the engine reports that the pair is unrated. It does not estimate what the rating would have been.

Engine: `scoreSocioCohort()` in `src/shared/socio-scoring.ts` – pure and dependency-free, so the Worker, the browser and the test suite all run the same code. Instrument definition: `src/shared/socio.ts`. Trend across rounds: `src/worker/lib/cohort-trend.ts`.